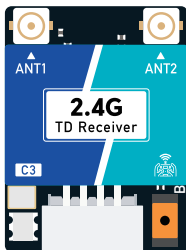




# ELRS Nano DUAL 2.4G True Diversity Receiver

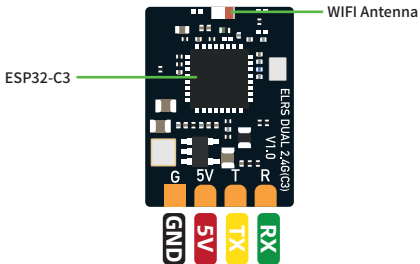
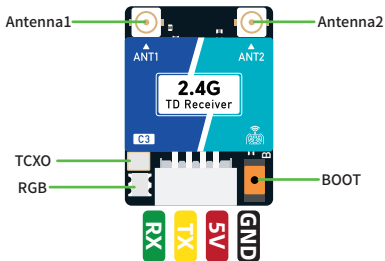
## User Manual



## Specifications

|                     |                                                   |
|---------------------|---------------------------------------------------|
| Model:              | GEPRC ELRS Nano DUAL 2.4G True Diversity Receiver |
| Firmware:           | GEPRC Nano True Diversity 2.4GHz RX               |
| Chips:              | ESP32-C3, SX1281x2                                |
| Dimension:          | 17*13mm                                           |
| Weight:             | 1.1g (RX only)                                    |
| Refresh Rate:       | 25Hz-1000Hz                                       |
| Crystal Oscillator: | TCXO                                              |
| Antenna Connector:  | ipex1                                             |
| TLM Power:          | 10mW                                              |
| Operating Voltage:  | 5V                                                |
| Frequency Band:     | 2.4GHz ISM                                        |
| Indicator Light:    | RGB light                                         |

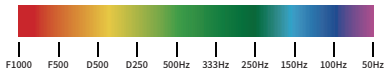
## Receiver diagram



## LED Status indication

| RGB Color | Status       | Description                                         |
|-----------|--------------|-----------------------------------------------------|
| Rainbow   | Fade effect  | Power on                                            |
| Green     | Quick flash  | WIFI upgrading mode                                 |
| Red       | Quick flash  | No RF chip detectde                                 |
| Orange    | Double flash | Binding mode                                        |
| Orange    | Triple flash | Connected, but mismatched model-match configuration |
| Orange    | Slow flash   | Waiting for connection                              |
|           | Solid on     | Connected and color indicates packet rate           |

### RGB color corresponding to 2.4G refresh rate:



F1000 and F500 are packet rates in FLRC mode, providing faster modulation and lower latency, but at the same time having shorter reception distance than normal Lora mode. This mode is great for racers.

D500 and D250 are packet rates in DVDA (Deja Vu Diversity Aid) mode. This mode works at the F1000 data packet rate of FLRC mode, providing better link connection in the case of complex interference by sending the same data packet multiple times. D500 and D250 indicate that the same data packet is sent twice and four times respectively.

# Instructions

General FC wiring diagram:



Betaflight standard connector wiring diagram:



Note: Since the Betaflight standard wiring sequence differs from the ELRS standard definition, please make sure to check that the connection order between the flight controller and receiver is correct when using the connector cable. Reversed polarity may cause damage to the receiver!

Open Betaflight Configurator, go to "Ports" tab and enable the corresponding UART as a Serial Rx (e.g. UART2 as shown below). Save and restart.

| Identifier | Configuration/MSP                            | Serial RX                           |
|------------|----------------------------------------------|-------------------------------------|
| USB VCP    | <input checked="" type="checkbox"/> 115200 ▼ | <input type="checkbox"/>            |
| UART1      | <input type="checkbox"/> 115200 ▼            | <input type="checkbox"/>            |
| UART2      | <input type="checkbox"/> 115200 ▼            | <input checked="" type="checkbox"/> |

On the "Configuration" tab, click on "Serial-based receiver" on the "Receiver" panel, and select "CRSF".

Receiver

Serial-based receiver(SPEKSAT,S ▾

Receiver Mode

**Note:** Remenber to configure a Serial Port(via Ports tab)and choose a se Receiver Provider when using RX\_SERIAL feature

CRSF ▾

Serial Receiver Provider

## Binding

The Receiver with firmware 3.5 or later, update the TX to the latest version before binding.

- 1.Power cycle the receiver three times (RGB light turn on before powering off) or hold the BOOT button for 7 seconds;
- 2.When the RGB light blink **orange**, it's in binding mode;
- 3.Use a 2.4G TX for binding.When the RGB light turns solid, binding is done.

## About ELRS

ExpressLRS project is being constantly updated - the contents of this manual cannot be kept up-to-date in time. For more information, please visit the ELRS Project official.

github: <https://github.com/ExpressLRS/ExpressLRS>

Official website: <https://www.expresslrs.org>

## Product list

- 1 x ELRS Nano DUAL 2.4G True Diversity Receiver
- 2 x 2.4G T-Antennas
- 1 x Heat shrink tube
- 1 x 4-pin silicone connect cable
- 1 x User manual

## Contact

Website: <https://geprc.com/>

**facebook**



[facebook.com/geprc](https://facebook.com/geprc)

Official  
website



[www.geprc.com](https://www.geprc.com)

**Instagram**



[instagram.com/geprc](https://instagram.com/geprc)

**You Tube**



**Manual**

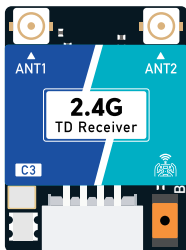


[geprc.com/support](https://geprc.com/support)



# ELRS Nano DUAL 2.4G 真分集接收机

使用说明书





## 基本参数

产品名称: GEPRC ELRS Nano DUAL 2.4G 真分集接收机

固件: GEPRC Nano True Diversity 2.4GHz RX

芯片: ESP32-C3, SX1281x2

尺寸: 17\*13mm

重量: 1.1g(仅接收机)

刷新率: 25Hz-1000Hz

晶振: 温补晶振

天线接口: ipex1

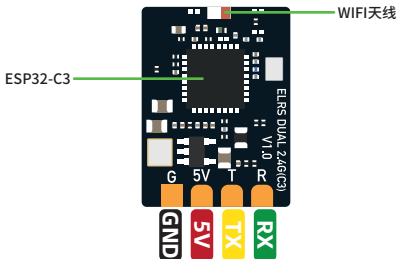
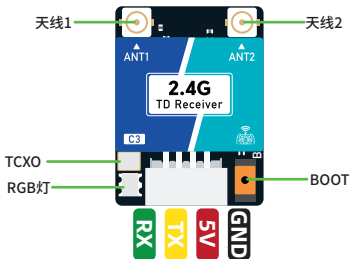
TLM功率: 10mW

工作电压: 5V

工作频段: 2.4GHz ISM

指示灯: 彩色RGB灯

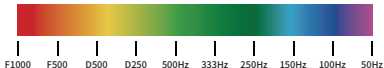
## 接收机示意图



## 状态灯含义

| 指示灯颜色 | LED灯状态 | 含义                 |
|-------|--------|--------------------|
| 彩虹    | 渐变     | 启动中                |
| 绿色    | 快闪     | WIFI模式             |
| 红色    | 快闪     | 未检测到射频芯片           |
| 橙色    | 双闪     | 对频状态               |
| 橙色    | 三闪     | 已对上频,但与模型匹配中的设置不一样 |
| 橙色    | 慢闪     | 无发射机信号             |
|       | 常亮     | 已连接,颜色代表不同的刷新率     |

### 2.4G刷新率对应的RGB颜色:



其中F1000和F500是工作在FLRC模式下的刷新率,相比LoRa模式,它可以提供更快的信号调制和更低的延时,但相比普通的LoRa模式接收距离会缩短,适合用于竞速飞行的场景;

D500和D250是工作在DVDA(Déjà Vu Diversity Aid)模式下的刷新率,DVDA模式是在FLRC模式1000Hz刷新率的基础上,通过多次发送相同的数据包,使得在环境干扰复杂的场景下也能提供更可靠的连接;D500和D250分别代表将相同的数据包发送两次和四次。

## 使用方法

常规飞控线序示意：



Betaflight标准连接器线序示意：



注意：由于Betaflight标准线序和ELRS标准线序定义不同，在使用端子线插接时，请务必检查飞控与接收机的连接线路是否正确，以免因为正负极接反而导致接收机损坏！

打开Betaflight地面站，转到端口界面，根据焊接情况，开启对应端口的“串行数字接收机”开关(以端口2为例)，然后保存重启。

| 标识符     | 设置/MSP                                       | 串行数字接收机                             |
|---------|----------------------------------------------|-------------------------------------|
| USB VCP | <input checked="" type="checkbox"/> 115200 ▼ | <input type="checkbox"/>            |
| UART1   | <input type="checkbox"/> 115200 ▼            | <input type="checkbox"/>            |
| UART2   | <input type="checkbox"/> 115200 ▼            | <input checked="" type="checkbox"/> |

设置接收机模式为“串行数字接收机”，并把串行数字接收机协议设置为“CRSF”。

|                                                 |           |
|-------------------------------------------------|-----------|
| 串行数字接收机(SPEKAT,SBU) ▼                           | 接收机模式     |
| <b>注意:</b> 使用串行接收机时, 请选择串行接收机类型, 并在串口页面设置相应的串口。 |           |
| CRSF ▼                                          | 串行数字接收机协议 |

## 对频操作

接收机出厂固件版本为3.5以上, 建议在对频前将高频头升级到最新版本。

- 1.接收机连续通断电三次(需要RGB灯亮起后才可断电), 或者通电后按住BOOT键持续7秒;
- 2.观察接收机RGB灯, 变为**橙色双闪**状态, 表示接收机已进入对频状态;
- 3.使用2.4G高频头进行对频,待接收机RGB灯由双闪变为慢闪, 再变为常亮状态, 即对频成功。

## 关于ELRS

由于ExpressLRS项目更新的速度比较快, 说明书中许多内容没法及时更新, 更多内容欢迎访问ELRS项目库和官方网站介绍。

github地址:<https://github.com/ExpressLRS/ExpressLRS>

官方网站:<https://www.expresslrs.org>

## 产品清单

1 x ELRS Nano DUAL 2.4G 真分集接收机

2 x 2.4G T型天线

1 x 热缩管

1 x 4pin 硅胶连接线

1 x 使用说明书

## 联系我们

格普官网:<https://geprc.com/>

格普淘宝:<https://geprc.taobao.com/>

格普官方 QQ 交流群:499699918